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Influence of thermophilic aerobic digestion as a sludge pre-treatment and solids retention time of mesophilic anaerobic digestion on the methane production, sludge digestion and microbial communities in a sequential digestion process

Hyun Min Jang [a], Hyun Uk Cho [a], Sang Kyu Park [b], Jeong Hyub Ha [a,b,**], Jong Moon Park [a,b,c,*]

[a] School of Environmental Science and Engineering, San 31, Hyoja-dong, Pohang 790-784, South Korea
[b] Department of Chemical Engineering, San 31, Hyoja-dong, Pohang 790-784, South Korea
[c] Division of Advanced Nuclear Engineering, Pohang University of Science and Technology, San 31, Hyoja-dong, Pohang 790-784, South Korea

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ABSTRACT

In this study, the changes in sludge reduction, methane production and microbial community structures in a process involving two-stage thermophilic aerobic digestion (TAD) and mesophilic anaerobic digestion (MAD) under different solid retention times (SRTs) between 10 and 40 days were investigated. The TAD reactor (R_{TAD}) was operated with a 1-day SRT and the MAD reactor (R_{MAD}) was operated at three different SRTs: 39, 19 and 9 days. For a comparison, control MAD ($R_{CONTROL}$) was operated at three different SRTs of 40, 20 and 10 days. Our results reveal that the sequential TAD–MAD process has about 42% higher methane production rate (MPR) and 15% higher TCOD removal than those of $R_{CONTROL}$ when the SRT decreased from 40 to 20 days. Denaturing gradient gel electrophoresis (DGGE) and real-time PCR results indicate that R_{MAD} maintained a more diverse bacteria and archaea population compared to $R_{CONTROL}$, due to the application of the biological TAD pre-treatment process. In R_{TAD}, *Ureibacillus thermophiles* and Bacterium thermus were the major contributors to the increase in soluble organic matter. In contrast, *Methanosaeta concilii*, a strictly aceticlastic methanogen, showed the highest population during the operation of overall SRTs in R_{MAD}. Interestingly, as the SRT decreased to 20 days, syntrophic VFA oxidizing bacteria, *Clostridium ultunense* sp., and a hydrogenotrophic methanogen, *Methanobacterium beijingense* were detected in R_{MAD} and $R_{CONTROL}$. Meanwhile, the proportion of archaea to total microbe in R_{MAD} and $R_{CONTROL}$ shows highest values of 10.5 and 6.5% at 20-d SRT operation, respectively. Collectively, these results demonstrate

* *Corresponding author*. Division of Advanced Nuclear Engineering, Pohang University of Science and Technology, San 31, Hyoja-dong, Pohang 790-784, South Korea. Tel.: +82 54 279 2275; fax: +82 54 279 8659.
** *Corresponding author*. School of Environmental Science and Engineering, San 31, Hyoja-dong, Pohang 790-784, South Korea. Tel.: +82 54 279 8315; fax: +82 54 279 8659.
E-mail addresses: jeonghha@postech.ac.kr (J.H. Ha), jmpark@postech.ac.kr (J.M. Park).
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that the increased COD removal and methane production at different SRTs in R_{MAD} might be attributed to the increased synergism among microbial species by improving the hydrolysis of the rate limiting step in sludge with the help of the biological TAD pre-treatment.

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1. Introduction

The activated sludge process is indisputably the most frequently employed technique in the municipal and industrial wastewater treatment plants (WWTPs). However, a significant amount of waste activated sludge (WAS) is produced in the activated sludge process, which poses problems with environment pollution. Most of sludge (~62%) in Korea had generally been discharged to the ocean before it was completely prohibited in 2012 by the London Convention 97 protocol (MOE, 2012). Even though sludge has been currently used for industrial application, co-firing feedstocks, and composting as a fertilizer, a large amount of WAS is still disposed of in landfills and by incineration (MOE, 2012). Disposals costs account for nearly 40–60% of the total WWTP operating costs (Appels et al., 2008). Negative effects of sludge disposal by landfill also include the serious contamination of land environments, with the destruction of habitat leading to subsequent loss of plant and animal species, since WAS generally contains pathogenic organisms, toxic organic substances and heavy metals, and inorganic nutrients such as phosphate and ammonium causing eutrophication (Campbell, 2000). In addition, sludge removal by incineration requires high energy consumption, giving rise to foul odor and generating the toxic chemicals that have a bad effect on human respiration (Keffala et al., 2013). Therefore, effective removal of WAS is a critical environmental challenge, especially with the recent stringency of environmental regulations and associated concerns (Paul et al., 2006).

Anaerobic digestion (AD) is generally perceived as a cost-effective alternative for the WAS treatment, because a large proportion of the organic matter can be converted to biogas (e.g., methane or hydrogen) or valuable products (e.g., organic acids) under anaerobic conditions (Li et al., 2011). Additionally, AD generates relatively low biomass, and the residues have potential uses as fertilizers and soil conditioners (Abubaker et al., 2012). However, without a pre-treatment step, the AD process has an organic removal efficiency of only 30–50% for solids retention time (SRT) of 20–40 days, because most of the WAS is composed of microbial cells enmeshed in extracellular polymeric substance (EPS), which is a sturdy structure against hydrolytic enzyme (Toreci et al., 2009). Hence, many types of pre-treatments have been developed to enhance the sludge solubilization, including mechanical, ultrasonic, chemical, thermal and combined thermo-chemical treatments (Carrère et al., 2010). A positive effect in terms of volatile solid (VS) destruction and methane production has been observed in previous studies that examined these pre-treatment methods. However, mechanical and thermal methods require a substantial amount of energy (Weemaes and Verstraete, 1998). Chemical treatment which is usually conducted with an acid or alkali, require large amount of chemicals to maintain the reaction conditions and neutralization after the reaction (Navia et al., 2002). In addition, inhibitory and biologically non-degradable compounds can be generated after thermal and chemical treatment (Stuckey and McCarty, 1984).

As an alternative treatment method, phase-separated digestion composed of two or more phases (e.g., mesophilic–mesophilic or thermophilic–mesophilic) has attracted attention recently. These types of digestion systems have various advantages compared to single-stage digestion, such as increased reactor stability, methanogen activity, and overall COD removal efficiency, since phase separation provides optimal growth conditions for bacteria and archaea group in each phase (Coelho et al., 2011). Among the digestion systems, temperature-phased anaerobic digestion (TPAD) processes have been widely applied and developed (Ge et al., 2011). Typically, TPAD consists of a thermophilic (45–60 °C) anaerobic process, followed by a mesophilic anaerobic process. The thermophilic phase is operated with a short SRT (<5 d) and a high organic loading rate (OLR) in order to accelerate hydrolysis and acidogenesis (Schmit and Ellis, 2001), while the mesophilic phase is operated with a relatively long SRT (>10 d) to obtain further hydrolysis and methanogenesis (Han et al., 1997). However, in such a phase configuration, one major drawback is the sensitivity of the thermophilic phase to the influent characteristics and OLRs (Song et al., 2004). In addition, operating the reactor in thermophilic anaerobic conditions increases the energy requirements (Ziemba and Peccia, 2011).

Some researchers have suggested combined anaerobic and aerobic processes in order to incorporate the advantages of aerobic and anaerobic digestion. In previous research on combined anaerobic and aerobic processes, with an SRT of 3–9 days, a higher VS reduction was achieved than in single-stage mesophilic or thermophilic anaerobic digestion (Kumar et al., 2006; Tomei et al., 2011). However, little attention has been paid to the applied aerobic step as a first stage in aerobic/anaerobic combined process. Particularly, very few studies have been reported with regard to thermophilic aerobic digestion (TAD) prior to a single mesophilic anaerobic digestion (MAD), which has achieved much higher solid reduction and methane production than single-stage MAD (Hasegawa et al., 2000; Pagilla et al., 2000). Based on these results, TAD might be applicable as the first stage for combined thermophilic aerobic/anaerobic sludge digestion processes, since TAD has a fast degradation rate of sludge leading to an increased amount of soluble organic products which are beneficial to methane production and to the self-heating ability, which can reduce the operational costs (Gomez et al., 2007).

To better understand and collect operating data related to the microbial community structure, numerous approaches have been developed and applied, including cultivation-dependent and independent approaches. PCR-based methods are used extensively, because they can detect both

cultivable and also non-cultivable bacteria (Nocker et al., 2007). Advanced PCR-based molecular approaches can provide qualitative and quantitative information about microbial community diversity and shifts in the process. Denaturing gradient gel electrophoresis (DGGE) is the most widely used for fingerprinting approaches, since it is less expensive than other fingerprinting methods, and it can provide taxonomic information (Gao and Tao, 2012). Real-time quantitative PCR (qPCR), which allows for the estimation of copy numbers of specific targeted genes in a sample, was used to quantify the bacteria and archaea group in the biological reactor (Shin et al., 2010b). The advantages of DGGE and qPCR facilitate the analysis of the relative and absolute microbial community. However, so far, the characterization of the microbial community structure and population dynamics in phase-separated digestion has been rarely explored, especially in combined TAD–MAD process.

In this study, a combined TAD–MAD process was examined to elucidate the effects of the TAD process as a biological pretreatment step on the methane production and sludge digestion in MAD under different SRTs, which inevitably affect the performance and microbial community structures. The feed sludge studied was predominantly WAS in that the substrate used consisted of 70% WAS and 30% primary sludge (v/v), since a primary sedimentation basin generating primary sludge has been widely employed in most of municipal wastewater treatment plants (WWPTs) and a proportion of primary sludge in the feed sludge for anaerobic digestion is about 20–40% (v/v) in Korea. Meanwhile, a culture-independent molecular approach involving DGGE and qPCR has been applied to the combined TAD–MAD process to investigate the microbial community structure and population dynamics.

2. Materials and methods

2.1. Feed sludge preparation

The feed sludge (Table 1) for this study was a mixture of primary and secondary sludge samples collected from the municipal WWTP in Daegu, Korea. In order to remove the particles which are larger than 1.0-mm (mainly inert materials), all sludge samples was filtered through a 1.0-mm sieve, and then the primary (TSS: 22.3 g/L; VSS: 18.4 g/L) and secondary (TSS: 62.8 g/L; VSS: 41.7 g/L) sludge samples (proportions 3:7 on v/v) were mixed thoroughly. The mixed sludge was transferred to 3-L bottles and stored at −25 °C until use.

2.2. Reactor setup and operating conditions

The sequential process (Fig. 1) consists of TAD (R_{TAD}) followed by MAD process (R_{MAD}). A single MAD process ($R_{CONTROL}$) is used as a control. Biological pretreatment occurs in R_{TAD}, and sludge digestion and CH_4 production occur in R_{MAD}. R_{TAD} was seeded with sludge (TSS: 48.2 g/L; VSS: 24.5 g/L) from a successfully-operated ATAD pilot plant in Daejeon, Korea. R_{MAD} and $R_{CONTROL}$ were seeded with mesophilic anaerobic sludge (TSS: 25.6 g/L; VSS: 14.8 g/L) from a treatment plant in Daegu, Korea. Before continuous operation, all reactors were operated in batch mode for two weeks. The reactors were fed four times per day, using a peristaltic pump (Cole–Parmer®) controlled by a timer and relay. During the experimental period, feeding and discharge were conducted simultaneously. Each reactor was operated as a CSTR with complete mixing, and the HRT and SRT in the system were equal. R_{TAD} was operated at SRT of 1 day, while R_{MAD} and $R_{CONTROL}$ were operated with three different SRTs ranging from 40 to 10 days over a period of 225 days.

Table 1 – Characteristics of feed sludge.

Parameters	Values (average ± standard deviation)
pH	6.61 ± 0.06
Alkalinity (g $CaCO_3$/L)	1.72 ± 0.06
TSS (g/L)	51.51 ± 5.22
VSS (g/L)	35.56 ± 2.47
TCOD (g/L)	71.35 ± 7.96
SCOD (g/L)	7.96 ± 0.8
Protein (g COD/L)	1.81 ± 0.29
Carbohydrate (g COD/L)	0.63 ± 0.13
Nitrogen	
TN (g/L)	4.51 ± 0.77
SN (g/L)	0.96 ± 0.12
NH_4^+–N (g/L)	0.90 ± 0.09
NO_2^- (g/L)	N.D.[a]
NO_3^- (g/L)	N.D.
Phosphorus	
TP (g/L)	1.81 ± 0.12
SP (g/L)	0.43 ± 0.02
PO_4^{3-}–P(g/L)	0.42 ± 0.01
Total VFAs (g COD/L)	1.54 ± 0.16
Acetic acid (g COD/L)	1.54 ± 0.16

a Not detected.

The R_{TAD} was operated at 55 ± 0.5 °C, and compressed air was continuously supplied at 5 L/min through a fine-pore diffuser to maintain aerobic conditions and excess air was directly vented to fume hood through an escape port. R_{MAD} and $R_{CONTROL}$ were operated at 35 ± 0.5 °C under strict anaerobic conditions. All reactor' pH was not controlled. More details about the experimental design and operating conditions are presented in Table 2.

2.3. Analytical methods

Samples were obtained from all reactors at intervals of about three days. The total suspended solids (TSS), volatile suspended solids (VSS), total chemical oxygen demand (TCOD), total alkalinity (TA), total nitrogen (TN), and total phosphorus (TP) were determined according to the procedures described in the Standard Methods (APHA–AWWA–WEF, 1998). For soluble COD (SCOD), ammonium (NH_4^+-N), soluble nitrogen (SN), and soluble phosphorus (SP) analysis, samples were filtered by a filter with 0.45 μm pore size. The carbohydrate and protein concentrations were measured using the phenol-sulfuric acid method (DuBois et al., 1956) and Lowry–Folin method (Lowry et al., 1951), respectively. Nitrite (NO_2^-), nitrate (NO_3^-), and orthophosphate (PO_4^{3-}–P) were determined by ion chromatography (ICS-1000, DIONEX Co., USA). The pH and oxidation reduction potential (ORP) in each reactor was continuously measured using a pH meter (405-DPAS-SC-K85, METTLER TOLLEDO, Switzerland) and ORP meter (Pt-4805,

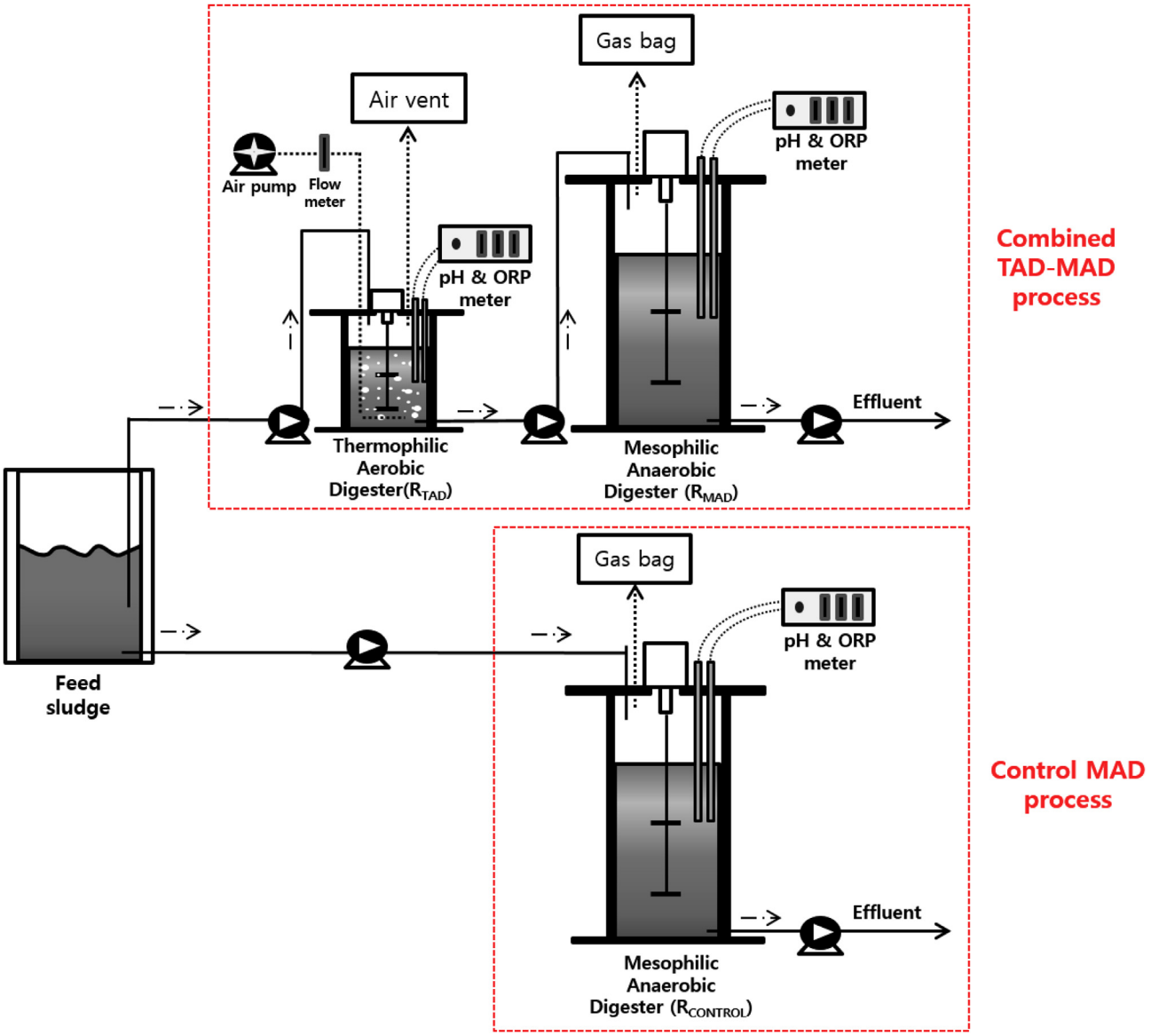


Fig. 1 – Schematic configuration of the lab-scale combined TAD–MAD process and control MAD process.

METTLER TOLLEDO, Switzerland). Biogas production from anaerobic reactors was quantified using a water displacement method, and detected using a gas chromatograph (Model 6890N, Agilent Inc., USA) equipped with a pulsed discharged detector (PDD). Volatile fatty acids (VFAs) were quantified by a high-performance liquid chromatograph (HPLC, Agilent Technology 1100 series, Agilent Inc., USA) equipped with a column (Aminex HPX-87H, BIORAD Inc., USA), a refractive

Table 2 – Experimental design of semi-continuous combined process and control MAD.

Parameters	TAD (R_{TAD})	MAD (R_{MAD})	Control MAD ($R_{CONTROL}$)
Reactor volume (L)	2	7	7
Working Volume (L)	0.125[a], 0.25[b], 0.5[c]	4.875[a], 4.75[b], 4.5[c]	5
Temperature (°C)	55 ± 0.5	35 ± 0.5	35 ± 0.5
Anaerobic process SRT (d)	–	39–9	40–10
Aerobic process SRT (d)	1	–	–
Air flow (L/min)	5	–	–

a 0–60 d.
b 60–132 d.
c 132–225 d.

index detector (RID), and a diode array detector (DAD). The quantified values including VFA, carbohydrate and protein content were converted to g COD/L using conversion factors (acetic acid: 1.066, carbohydrate: 1.07, protein: 1.50).

2.4. Protease activity test

To investigate the key enzyme activity related to the hydrolysis, protease activity was measured since it is considered as one of the main enzymes under both thermophilic aerobic and mesophilic anaerobic sludge digestion (Yan et al., 2008; Yang et al., 2010). Protease activity was measured by modified Anson's method (Anson, 1938) using casein as substrate (Cupp-Enyard, 2008). 5 mL of casein solution (0.65% w/v in 50 mM potassium phosphate buffer, pH 7.5) and 1 mL of sample were added to a vial. The incubation was carried out at 55 °C (for R_{TAD}) and 35 °C (for R_{MAD} and $R_{CONTROL}$) for exactly 10 min, respectively. After 10 min reaction, 5 mL of the 110 mM trichloroacetic acid was added to stop the reaction. Then, the solutions incubated at 37 °C for 30 min to stabilize the reaction. After the 30 min incubation, solutions were filtered by 0.45 μm pore size syringe filter to remove any insoluble particle. Then, 2 mL of filtrates were mixed with 1 mL of 2 N Folin's reagent and 5 mL of 500 mM sodium carbonate solution to buffer any pH drop which can be caused by addition of the Folin's reagent. Mixed solutions incubated at 37 °C for 30 min. After this incubation, solutions were filtered by 0.45 μm pore size syringe filter into cuvettes and absorbance was measured at 660 nm wavelength with tyrosine standard. One unit of protease activity (Unit) was defined as the amount of protease that can hydrolysis casein to 1 μg of tyrosine for 1 min at 55 °C (for R_{TAD}) and 35 °C (for R_{MAD} and $R_{CONTROL}$), respectively.

2.5. Molecular microbial analysis

2.5.1. Genomic DNA extraction

After sampling at the startup and upon the achievement of steady state in each phase, 1 mL of sample was centrifuged twice at 13,000 g for 5 min, and sludge pellets were washed in deionized distilled water (DDW). The washed sample was re-suspended in 200 μL of DDW. Total DNA extraction and purification was conducted using a Nucleo Spin® Soil kit (MACHEREY-NAGEL, Germany). The purified DNA was eluted with 50 μL of 1 × TE buffer and stored at −25 °C for further analysis.

2.5.2. PCR-DGGE analysis

PCR-based community fingerprinting techniques, particularly PCR-DGGE, are widely used to investigate the difference or change of microbial community structures in a bioreactor. Domain-level PCR was conducted using a My Cycler™ Personal Thermal Cycler (Bio-Rad Corp., USA) with universal primers (Table 3) (Shin et al., 2010b). 50 μL of the PCR reaction mixture was prepared using a *TaKaRa Ex Taq* polymerase kit (TaKaRa Code: RR001A, TaKaRa BIO INC., Japan), with 5 μL of 10 × PCR Buffer, 4 μL of dNTP (2.5 mM each), 2 μL each primer (final concentration 1 μM), 0.25 μL of *Ex Taq* polymerase (1.25 U), 4 μL of template and 32.75 μL of sterile distilled water. The PCR protocol was conducted with (1) an initial denaturation at 95 °C for 10 min; (2) 30 cycles of 95 °C for 5 min, 55 °C for 30 s, and 72 °C for 30 s; and (3) a final extension at 72 °C for 10 min. DGGE profiling was conducted using a Dcode™ Universal Mutation Detection System (Bio-Rad Corp., USA). The PCR product was loaded onto each well of an 8% (w/v) acrylamide gel (acrylamide: bisacrylamide solution, 37.5:1) containing a 30–60% denaturant gradient, where 100% denaturant agent was defined as 7 M urea with 40% formamide. Electrophoresis was performed in 0.5× TAE buffer for 720 min at 100 V and 60 °C. After electrophoresis, gel was stained with 0.5 × TAE buffer containing SYBR green I nucleic acid gel stain (1:10,000 dilution, FMC BioProducts, USA) for 20 min and then de-stained for 20 min with 0.5 × TAE buffer. Then the visualized gel was photographed using a Gel Doc™ Imaging System (Bio-Rad Corp., USA).

All visual bands were excised directly from the DGGE gel and washed by 100 μL of sterile distilled water. After washing, excised bands were eluted with 30 μL of TE buffer at 4 °C for 48 h. Each band eluted solution was amplified with corresponding primers without the GC-clamp (Table 3.). The PCR products were cloned using pGEM-T Easy Vector (Promega Corp., USA), and the cloned 16S rRNA gene fragments were sequenced (SolGent Co., Ltd., Korea). The closest phylogenetic

Table 3 – Primer sets for the DGGE and qPCR (Shin et al., 2010b).

Primer set	Target group	Sequence (5′ –3′)	Representative strains (culture collection)	Linear range (copy/μL)
BAC338F BAC805R	Bacteria[a]	ACTCC TACGG GAGG CAG GACTA CCAGG GTATC TAATC C	*Escherichia coli* K12 (DSM 1607)	2.5×10^3–3.5×10^9 (0.995)[b]
ARC787F ARC1059R	Archaea[a]	ATTAG ATACC CSBGT AGTCC GCCAT GCACC WCCTC T	*Methanomicrobium* mobile BP (DSM 1539)	2.0×10^3–4.2×10^9 (0.999)
MBT857F MBT1196R	Methanobacteriales	CGWAG GGAAG CTGTT AAGT TACCG TCGTC CACTC CTT	*Methanobacterium formicicum* M.o.H. (DSM 863)	3.6×10^2–3.8×10^8 (0.996)
MMB282F MMB832R	Methanomicrobiales	ATCGR TACGG GTTGT GGG CACCT AACGC RCATH GTTTA C	*Methanomicrobium* mobile BP (DSM 1539)	1.0×10^3–2.9×10^9 (0.998)
MCC495F MCC832R	Methanococcales	TAAGG GCTGG GCAAG T CACCT AGTYC GCARA GTTTA	*Methanococcus voltae* (DSM 1537)	6.1×10^2–5.5×10^8 (0.998)
MSL812F MSL1159R	Methanosarcinales	GTAAA CGATR YTCGC TAGGT GGTCC CCACA GWGTA CC	*Methanosarcina barkeri* MS (DSM 800)	7.2×10^2–8.9×10^8 (0.993)

a When performing DGGE analysis, a 40-bp GC-clamp was added at the 5′ ends of BAC338F and ARC787F.
b r^2 of slope.

affiliation of the sequence from DGGE gels were compared with the reference using the National Centre for Biotechnology Information (NCBI) BLAST program.

2.5.3. *qPCR analysis*

To investigate the relationship between the domain-level group and four major methanogen orders (Methanobacteriales, Methanococcales, Methanomicrobiales, and Methanosarcinales) with regard to population and reactor performance, qPCR amplification and fluorescence detection were performed using an Applied Biosystems 7300 qPCR system (Applied Biosystems, Forster City, USA). 20 μL of the qPCR reaction mixture was prepared: 10 μL of TaKaRa SYBR® Premix Ex Taq™ (TaKaRa BIO INC., Japan), 0.4 μL of each primer (0.1 μM for bacteria, 0.2 μM for archaea and methanogens), 0.4 μL of 50 × ROX reference dye, 20 ng of template DNA, and 6.8 μL of PCR-grade water. Each primer set was designed in accordance with previous research (Table 3) (Shin et al., 2010b). The primer concentration and PCR conditions were separately optimized for each primer set. The qPCR step for quantification was as follows: 95 °C for 10 s, followed by 40 cycles of 95 °C for 5 s, 59 °C for 10 s (56 °C for the total archaea), 72 °C for 27 s (fluorescence detection step).

10-fold serial diluted (10^1–10^9 copies) 16S rRNA from the representative strains (Table 3) was used to construct the standard curve for each primer sets as described previous research (Lee et al., 2008). The concentrations of the nucleic acid were measured in triplicate using a spectrophotometer equipped with a Nano cell (MECASYS Inc., Korea), and the 16s rRNA gene copy concentration was calculated according to an equation described in previous research (Whelan et al., 2003). The threshold cycle (Ct) value was automatically calculated using the SDS software of the 7300 qPCR system (Applied Biosystems, Forster City, USA). The Ct values were plotted against the logarithm of their template copy concentrations, and the 16S rRNA gene copy concentrations of target groups were determined against the corresponding standard curves within the linear range ($r^2 > 0.993$). All amplifications were performed in duplicate with non-template control (NTC).

3. Results and discussion

3.1. *The process stability: pH, TA, VFA and ORP*

To operate the process in a stable condition, the reactor operation was started with the longest total SRT, which was 40 days. As described in Section 2.2, R_{TAD} was operated as a biological pre-treatment process with an SRT of 1 day during the overall experiment period, which previous experimental results have shown to be the optimum SRT for reducing sludge and producing soluble products for methane production (data not shown). R_{MAD} and $R_{CONTROL}$ were operated at three different SRTs: 39, 19, 9 and 40, 20, 10 days at corresponding organic loading rates (OLRs) of 1.84, 3.75 and 7.25 kg COD/m^3 d (0.81, 1.61 and 3.23 kg VSS/m^3 d), respectively in order to examine the effects of the SRT of the mesophilic anaerobic digestion process on sludge removal, methane production and microbial communities in combined process.

Fig. 2 shows the pH, TA, VFA and ORP values in all reactors for 225 days. Typically, pH variation in the anaerobic process is highly linked with VFA accumulations, and its decline to a certain level can be an inhibition factor for methane production (Jun et al., 2009). In this study, no considerable accumulation of VFA in R_{MAD} and $R_{CONTROL}$ were observed, and the pH ranges in both anaerobic reactors were between 7.0 and 7.5 until the 20-d SRT. However, a distinct decline of pH in R_{MAD} and $R_{CONTROL}$ was detected, which resulted from the accumulation of metabolic intermediary products, especially VFAs, with a 10-d SRT. The results suggested that the 10-d SRT was not feasible for maintaining the stability of the process. Unlike R_{MAD} and $R_{CONTROL}$, R_{TAD} maintained a relatively high pH (7.9–8.2) during the overall digestion due to the higher aeration rate and thermophilic conditions, which caused ammonia and CO_2 stripping.

The TA has been considered to have a buffering capacity in AD, since the sudden pH drop resulting from the accumulation of VFA could be alleviated by the alkalinity in the AD (Björnsson et al., 2001). As shown in Fig. 2b, TAs in the R_{MAD} and $R_{CONTROL}$ were higher than 2.9 g $CaCO_3$/L until the 20-d SRT. This was due to the archaea and bacteria species

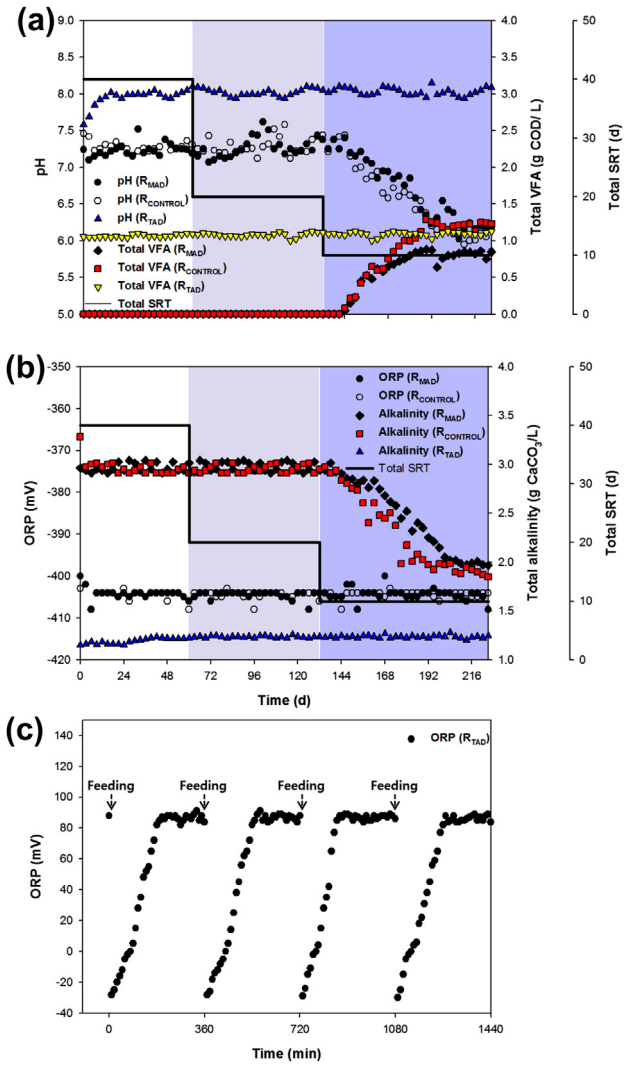


Fig. 2 – Change in reactor stability parameters during the overall digestion: (a) pH and Total VFA, (b) ORP and Total alkalinity, (c) ORP variation patterns in R_{TAD} after feeding.

producing CO_2, HCO_3^-, and NH_3 (Appels et al., 2008). Also, higher ammonium concentrations in the MAD likely led to higher alkalinity concentrations compared to those of the influent (data not shown). As with the pH, TAs in the R_{MAD} and $R_{CONTROL}$ continually decreased to 1.97 and 1.85 g $CaCO_3$/L at 10-d SRT, respectively. On the other hand, TA in R_{TAD} was quite low and stable in the range of 1.15–1.25 g $CaCO_3$/L during the digestion, since continuous aeration led to ammonia and CO_2 stripping.

In the biological process, the ORP measurement can be a useful indicator for monitoring the reactor conditions because it shows not only the aerobic but also anaerobic conditions. During the overall digestion, R_{MAD} and $R_{CONTROL}$ show no ORP change, and the values were lower than − 400 mV (Fig. 2b). These results indicated that they were maintained under strict anaerobic conditions. In contrast, the ORP value in R_{TAD} varied from −25 to 90 mV, because the feed sludge was semi-continuously provided, although air was continually supplied. After feeding, it sharply declined to −25 mV, and then recovered to around 90 mV within 3 h. This result is also supported by a previous study, which reported the same ORP variation patterns in semi-continuous combined mesophilic aerobic and anaerobic conditions (Novak et al., 2011).

3.2. *VSS and COD reduction*

To evaluate the effect of SRT on the solid removal efficiency in a combined process, the VSS removal was studies, which is a significant solid indicator in the biological process (Yang et al., 2010). Fig. 3 shows the average VSS reduction (VSSR) efficiency of the combined TAD-MAD and $R_{CONTROL}$ in steady state at each SRT. With the 40-d SRT, the VSSR in $R_{CONTROL}$ was highest at 45 (VSS decreased from 35.56 to 19.56 g/L) %. As the SRT decreased from 20 to 10 days, the VSSR in $R_{CONTROL}$ decreased from 41 (VSS decreased from 35.56 to 20.98 g/L) to 27.5 (VSS decreased from 35.56 to 25.78 g/L) %. In contrast,

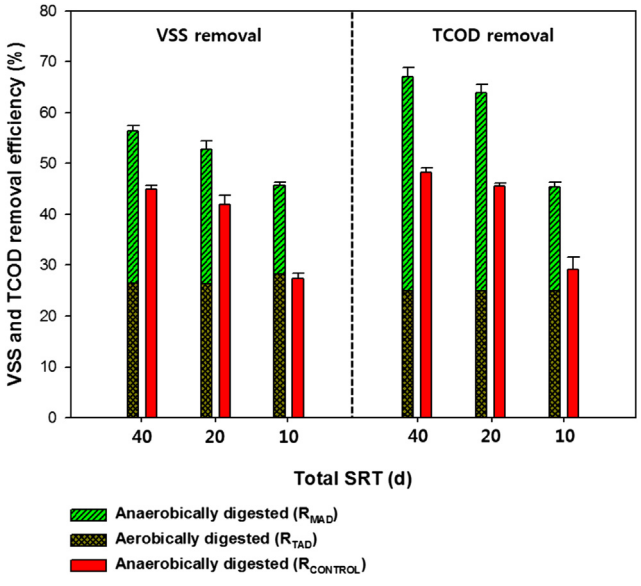


Fig. 3 – VSS and TCOD removal efficiency (%) in the combined TAD–MAD process and control MAD process with the SRTs.

although the total SRT decreased to 10 days, the combined TAD–MAD process achieved higher VSSR than 45% with the help of TAD pre-treatment. The combined process achieved VSSRs of 57 (VSS decreased from 35.56 to 15.29 g/L), 51 (VSS decreased from 35.56 to 17.42 g/L) and 45 (VSS decreased from 35.56 to 19.56 g/L) % at SRTs of 40, 20, and 10 days, respectively. This higher VSSR in the combined TAD-MAD could be attributed to the high exogenous and endogenous activities in R_{TAD} (LaPara and Alleman, 1999). Even though R_{TAD} was operated with 1-d SRT, it shows a VSSR of about 27 (VSS decreased from 35.56 to 25.96 g/L) % during the overall digestion. In addition, high endogenous decay might lead to low net biomass production in the R_{TAD} (Abeynayaka and Visvanathan, 2011). These differences in VSSR between the $R_{CONTROL}$ and the combined process demonstrate the advantages of TAD as a pre-treatment stage in the sludge digestion system.

Fig. 3 shows that the average total chemical oxygen demand (TCOD) removal efficiencies in $R_{CONTROL}$ were 48, 45, and 28% at SRTs of 40, 20, and 10 days, respectively. In contrast, the values in combined TAD-MAD were 67, 63, and 45% at SRTs 40, 20, and 10 days, respectively. The decreasing SRT causes a decrease of TCOD removal efficiency, and a significant decline were observed with the 10-d SRT in both processes. This result can be attributed to the fact that the 10-d SRT is insufficient to maintain the activated anaerobic bacteria and archaea populations. Although the TCOD removal efficiency decreased while the SRT decreased, the combined process shows about 15% higher TCOD removal efficiency than $R_{CONTROL}$ during the overall digestion. These results indicate that the biodegradability of sludge was enhanced with the TAD pre-treatment in the combined process.

3.3. *Soluble organic materials (SOM) and protease activity*

Table 4 summarizes the concentrations of SOM and protease activity in all reactors. The data included in this table represent the average values once a steady state was reached at each SRT. In R_{TAD}, the SCOD representing the total SOM value increased from 7.96 to 12.75 g/L due to the high amount of VSS destruction by thermophilic aerobic bacteria, otherwise approximately 85% of the SOM produced from R_{TAD} was consumed and converted into methane by anaerobic microbial species in R_{MAD} with the 40-d and 20-d SRT. Also, $R_{CONTROL}$ shows a high consumption of SOM at these SRTs. The sum of soluble carbohydrate and protein accounts for more than 60% of the SOM in R_{MAD} and $R_{CONTROL}$ during the overall digestion. In general, the carbohydrate and protein in the anaerobic process are known to be easily and rapidly converted to simple sugars, amino acids, and further degraded to VFAs (McInerney, 1988). As shown in Table 4, the protein concentration is higher than that of carbohydrate in R_{MAD} and $R_{CONTROL}$ at all SRTs. Fang and Yu (2000) and Lee et al. (2009) observed low protein biodegradability when the concentrations of carbohydrate and protein were both high. This phenomenon was explained by the high concentration of carbohydrate inhibiting the synthesis of extracellular proteolytic enzymes, which is highly related to protein hydrolysis (McInerney, 1988). In addition, protein has a lower digestion rate than carbohydrate (Gujer and Zehnder, 1983). As the SRT decreased from 20 to 10 days,

Table 4 – Characteristics of soluble organic materials (SOMs) and protease activities from each reactor at different solid retention times (Average ± standard deviation).

Parameter	TAD (R_{TAD})	MAD (R_{MAD})			Control MAD ($R_{CONTROL}$)		
SRT	1 d	39 d	19 d	9 d	40 d	20 d	10 d
SCOD (g/L)	12.75 ± 0.14	2.04 ± 0.04	2.14 ± 0.05	5.73 ± 0.08	1.85 ± 0.12	2.14 ± 0.09	6.24 ± 0.07
Carbohydrate (g COD/L)	2.96 ± 0.64	0.63 ± 0.01	0.71 ± 0.08	2.16 ± 0.08	0.46 ± 0.01	0.69 ± 0.03	1.95 ± 0.04
Protein (g COD/L)	2.16 ± 0.38	0.89 ± 0.01	1.01 ± 0.01	2.56 ± 0.08	0.83 ± 0.05	0.93 ± 0.06	3.12 ± 0.08
Total VFA[a] (g COD/L)	1.06 ± 0.24	–	–	0.85 ± 0.02	–	–	1.23 ± 0.04
Protease activity (Unit/mL)	0.82 ± 0.02	0.35 ± 0.04	0.36 ± 0.03	0.10 ± 0.01	0.28 ± 0.02	0.25 ± 0.02	0.08 ± 0.01

a Only acetic acid was detected in this study.

a large proportion of the SOM was discharged in both anaerobic processes, owing to the shorter SRT and overloaded SOM, which led to a serious accumulation of VFAs, resulting in a drop of the pH in R_{MAD} and $R_{CONTROL}$ to 6.15 and 5.94, respectively (Fig. 2a). In addition, according to previous results, the accumulated VFAs can induce local pH variation around hydrolysable biomass surfaces, which inhibits the process performance (Vavilin et al., 2008).

As shown in Table 4, protease activity in R_{TAD} shows constant and high value (0.82 Unit/mL), although it has relatively high carbohydrate concentration (2.96 g COD/L) and short SRT (1-day). This is highly matched with relatively higher VSSR efficiency in R_{TAD} (Fig. 3), and no significant protease activity inhibition was observed during the overall digestion. As like R_{TAD}, both anaerobic reactors show relatively constant protease activities (0.35–0.36 and 0.25–0.28 Unit/mL in R_{MAD} and $R_{CONTROL}$, respectively) during 40- and 20-d SRT operations. In addition, R_{MAD} shows higher protease activities than that of $R_{CONTROL}$ during the overall digestion. This result indicated that TAD pre-treatment stage also enhanced the enzyme activities performing hydrolysis in the following reactor (R_{MAD}). With further decrease of SRTs from 20 to 10 days, significant decline of protease activities was observed in both anaerobic reactors. This trend is also highly consistent with the decline of VSSR efficiency at the 10-d SRT in both processes.

3.4. *Methane production*

To investigate the relationship between SRT and methane production in the combined TAD–MAD, the methane production was recorded continuously during the overall digestion. Fig. 4a shows the methane production rate (MPR) for both processes during the digestion. Even with the same variation patterns of MPR, the MPR in the combined process was consistently higher than that of $R_{CONTROL}$. By applying R_{TAD}, an increment in MPR of approximately 42% was achieved. Both processes experienced significant increases in MPR when the SRT decreased from 40 to 20 days. The MPR increased from 152 to 261 mL CH_4/L/d in the combined process and from 110 to 185 mL CH_4/L/d in $R_{CONTROL}$. However, a marked MPR decline was observed for both processes with a 10-d SRT.

The methane yield is shown in Fig. 4b in terms of the milliliters of methane produced per gram of VSS removed and the methane content in the produced biogas for both processes at each SRT. The average methane yield was relatively consistent with the 40-d and 20-d SRTs, with values in the ranges of 288–300 and 253–272 mL CH_4/g VSS removed in the combined process and control $R_{CONTROL}$, respectively. With regard to the methane content, the combined process shows the highest value in the range of 70.8–72.4% compared to 63.5–65.5% in the $R_{CONTROL}$ at the 40-d and 20-d SRTs. In general, 48–65% methane content in biogas is considered a typical value for AD (Ward et al., 2008). This higher methane content in R_{MAD} might be partially due to the greater amount of CO_2 being dissolved in the form of HCO_3^- or CO_3^{2-}, which can be used as the substrate for hydrogenotrophic methanogens. As the SRT decreased to 10 days, the methane yield and content were instantaneously decreased in both processes. Based on the results, it was speculated that operation with a 20-d SRT

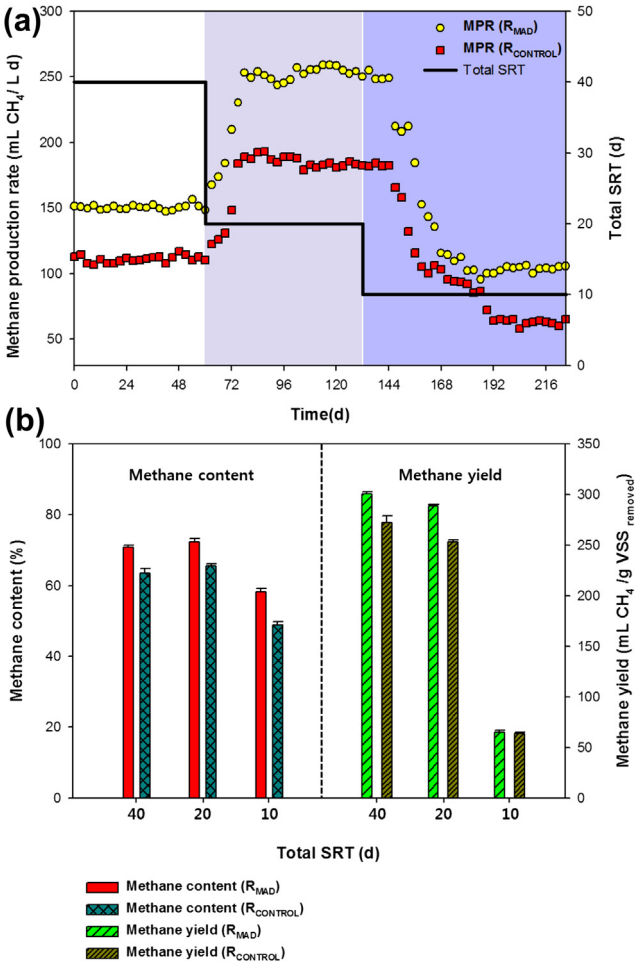


Fig. 4 – Variation of (a) methane production rate (MPR), (b) methane content and yield with the SRTs.

appeared to favor the methanogenic performance, resulting in higher MPR compared to that of the 40-d and 10-d SRTs.

3.5. Microbial community structure analysis

3.5.1. Qualitative analysis by PCR-DGGE

In order to evaluate the microbial community structure and relative shifts in abundance during the SRT changes, the PCR products amplified with a bacteria and archaea primer set were analyzed using DGGE analysis. All of the 16S rRNA gene sequences retrieved from DGGE bands show a high degree of similarity (above 90%) to the reference in the NCBI database (Table 5).

As shown in Fig. 5a, totals of 7, 10, and 6 bacteria DGGE bands, designated as T1-7, M1-10, and C1-6, were detected in R_{TAD}, R_{MAD}, and $R_{CONTROL}$, respectively. Although R_{TAD} was seeded with TAD sludge taken from an ATAD pilot plant, bands T2 and T3 disappeared, since the bacterial community in TAD can be shifted by various digestion conditions (e.g., reactor type, oxygen level and OLR) (Piterina et al., 2012). On the other hand, some new bands (T1, T4, T5 and T6) appeared throughout the entire digestion. Bands T4 and T5 were assigned to *Petrobacter* sp. (>90%), which is an aerobic and moderately thermophilic bacterium (Salinas et al., 2004). The prevailing band intensities (T1, T6, and T7) were observed during the overall digestion. Sequences recovered from T1 and T7 were closely affiliated with *Ureibacillus thermophiles* and Bacterium thermus with 99 and 92 % similarity, respectively. According to previous reports, *U. thermophiles* and Bacterium thermus were mainly detected in the thermophilic aerobic sludge digestion process. These species can produce thermoactive hydrolytic extracellular enzymes (e.g., protease, lipase and cellulase) (Chen et al., 2004; Ugwuanyi et al., 2008; Liu et al., 2010). In addition, the relatively high protease activity in R_{TAD} (Table 4) was consistent with appearance of these species. This indicates that the bacteria related to the T1 and T7 bands might be highly responsible for the hydrolysis of sludge in R_{TAD}. T6, one of the major bands in R_{TAD}, shows 98% similarity to *Clostridium straminisolvens*. The presence of anaerobic cellulolytic *Clostridia* spp. within the TAD has been reported in previous research, but whether those species maintain their activity in TAD is still unclear (Piterina et al., 2012). In this study, a high band intensity of T6 suggests that *C. straminisolvens* plays a major role in R_{TAD}. In addition, the oxygen and temperature tolerance of this species supports the potential activities of such species in R_{TAD} (Kato et al., 2004).

There were apparent differences in the bacterial community structure between R_{TAD} and both R_{MAD} and $R_{CONTROL}$ anaerobic reactors (Fig. 5a). Even though the SRTs were decreased from 40 to 10 days, some bands were detected

Table 5 – Sequence affiliation and putative function of bands obtained from PCR-DGGE.

Band name	Nearest sequence in database	Similarity (%)	Phylogenetic group	Putative function	Accession number
Bacteria					
T1	*Ureibacillus thermophilus*	99	Firmicutes	Hydrolysis	AB682456
T2	Uncultured actinobacterium clone HV9	95	Actinobacteria	Unknown	JF303826
T3	*Coprothermobacter* sp.	98	Firmicutes	Hydrolysis	AB537980
T4	*Petrobacter* sp.	90	β-Proteobacteria	Hydrolysis	DQ539621
T5	*Petrobacter succinatimandens* sp.	100	β-Proteobacteria	Acid oxidizing	NR025725
T6, M10	*Clostridium straminisolvens*	98	Firmicutes	Hydrolysis	NR024829
T7	Bacterium thermus-lsg2	92	β-Proteobacteria	Hydrolysis	HQ436531
M1, C1	Uncultured *Symbiobacterium* sp.	99	Firmicutes	Hydrolysis	AB052397
M2, C2	Uncultured gamma proteobacterium	95	γ-Proteobacteria	Acidogenesis	FM252968
M3	*Aeriscardovia aeriphila*	91	Actinobacteria	Hydrolysis	NR042759
M4	Uncultured bacterium	95	Unknown	Unknown	FN435252
M5, C3	*Clostridium ultunense* sp.	99	Firmicutes	Acid oxidizing	NR026531
M6, C4	*Bacillus* sp.	94	Firmicutes	Acidogenesis	AJ489379
M7, C5	*Lactobacillus amylovorus*	95	Firmicutes	Acidogenesis	NR043287
M8	*Streptomyces* sp.	98	Actinobacteria	Hydrolysis	AM889490
M9, C6	*Nocardioides* sp.	95	Actinobacteria	Hydrolysis	AB508351
Archaea					
A1	*Methanosarcina* sp. MC-15	99	Methanosarcinales	Aceticlastic	JF812257
A2, A12	Uncultured *Methanogenium* sp.	93	Methanomicrobiales	Hydrogenotrophic	JN173203
A3, A13	Uncultured Methanomicrobiales	98	Methanomicrobiales	Hydrogenotrophic	FR836472
A4	*Methanosarcina siciliae*	99	Methanosarcinales	Aceticlastic	U89773
A5, A14	*Methanospirillum hungatei* strain GRAU-3	99	Methanomicrobiales	Hydrogenotrophic	JN004141
A6, A15	*Methanosarcina siciliae*	99	Methanosarcinales	Aceticlastic	FR733698
A7, A16	*Methanosaeta concilii*	99	Methanosarcinales	Aceticlastic	AB679168
A8, A17	*Methanolinea tarda* NOBI-1	99	Methanomicrobiales	Hydrogenotrophic	NR028163
A9, A18	Uncultured Methanosarcinales	99	Methanosarcinales	Aceticlastic	CU917434
A10, A19	*Methanobacterium beijingense*	100	Methanobacteriales	Hydrogenotrophic	AY350742
A11, A20	*Methanoculleus* sp. Annu8	98	Methanomicrobiales	Hydrogenotrophic	HM630582

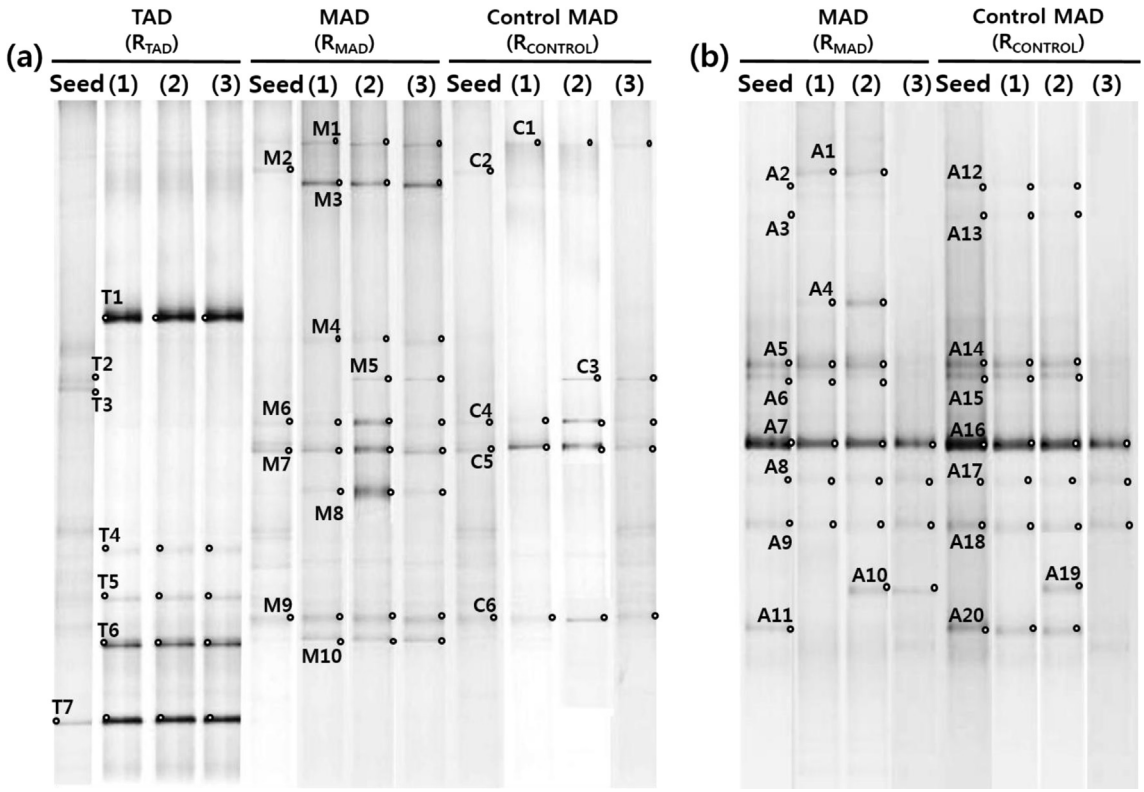


Fig. 5 – DGGE profiles of the PCR products amplified with universal primer: (a) bacteria profiles, (b) archaea profiles. (1):60 d, (2):120 d, (3):216 d.

throughout the digestion in the seed sludge that are closely related to *Bacillus* sp. (94%) (M6 and C4) and *Lactobacillus amylovorus* (95%) (M7 and C5), which are mainly involved in acidogenesis and acetogenesis (Nakamura, 1981), coupled with aceticlastic methanogens. Shin et al. (2010a) reported that these species are predominant acidogens in acidogenic reactors with short HRTs (<4.2 day). In addition, M9 and C6 closely affiliated with *Nocardioides* sp. (95%) which belongs to Actinobacteria also were detected in both anaerobic reactors during the overall digestion. This species is known as a producer of hydrolytic enzyme and is frequently detected in anaerobic digestion environments (Rintala and Puhakka, 1994; Levén et al., 2007). Although M1 and C1, which are closely related to Uncultured *Symbiobacterium* sp. (99%), were not detected in the seed sludge, this species was also detected in both anaerobic reactors through the digestion. The appearance of this species might enhance the activities of *Bacillus* sp., because they maintain symbiotic interactions with different species, especially *Bacillus* sp. (Ueda et al., 2004).

Even though the bacteria DGGE band profiles were similar, there were discernible distinctions observed between R_{MAD} and $R_{CONTROL}$. Interestingly, *Aeriscardovia aeriphila* (91%) (M3) and *Streptomyces* sp. (98%) (M8), belong to Actinobacteria were detected only in R_{MAD} (Fig. 5a). These species are known to have the ability to form hydrolytic enzyme or organic acid in a methanogenic reactor (Simpson et al., 2004; Supaphol et al., 2011). Furthermore, M10 (T6), closely affiliated with *C. staminisolvens* (98%), was detected in both R_{TAD} and R_{MAD}. Under anaerobic conditions, it can produce organic acid using various SOMs as well as cellulolytic enzyme (Kato et al., 2004). These changes in community structure in R_{MAD} indicate that the appearance of these additional species was undoubtedly affected by the TAD process, and might be highly related to enhanced reactor performance in R_{MAD}.

Another notable finding observed in the bacteria DGGE was the syntrophic VFA oxidizing bacteria, *Clostridium ultunense* sp. (99%) (M5 and C3), coupled with hydrogenotrophic methanogens, which were detected as the SRT decreased to 20 days. Schnürer et al. (1996) detected *C. ultunense* sp. in mesophilic anaerobic conditions, and reported that they can convert VFAs into H_2 and CO_2 before they were directly converted to methane by aceticlastic methanogens. The band fluorescence intensity related to these bacteria still remained with 10-d SRT operation, which is expected because the wide pH range (5–10) for growth was attributed to the maintenance of the population, even though an abrupt pH drop occurred in the R_{MAD} and $R_{CONTROL}$ (Fig. 2a)(Schnürer et al., 1996).

Fig. 5b shows the archaea DGGE profiles in R_{MAD} and $R_{CONTROL}$ during the digestion. A total of 20 bands (A1-20) were detected, and all of the sequences retrieved from those bands were involved in three methanogen orders: Methanosarcinales (MSL), Methanomicrobiales (MMB) and Methanobacteriales (MBT) (Table 5).

Bands A7 and A16 showed the highest band fluorescence intensity during the digestion. The sequence recovered from these bands was closely related to *Methanosaeta concilii* (99%),

which is known as an aceticlastic methanogen. This indicates that the methanogens corresponding to this band are mainly responsible for methane production in both anaerobic reactors. As the seed sludge in R_{MAD} experienced anaerobic digestion, bands A2, A3, and A11, related to hydrogenotrophic methanogens disappeared. On the other hand, band A1 and A4, which highly related to aceticlastic methanogens, appeared until 20-d SRT operation. The high SOM concentration resulting from the TAD pretreatment might have an influence on the archaea community structure in R_{MAD}.

As the SRT decreased to 20 days, bands A10 and A19, closely related to *Methanobacterium beijingense* (100%), the only species affiliated with the order MBT, were detectable in both anaerobic reactors. The appearance of these bands precisely coincided with the appearance of *C. ultunense* sp. (Fig. 5a), because this species is highly coupled with hydrogenotrophic methanogens. The reason for the appearance of the A10 band rather than the recovery of the bands A2, A3, and A11 in R_{MAD} was unclear. It is speculated that the different growth rates and behavior between MMB and MBT might contribute to the appearance of A10.

Although sufficient SOM (i.e., a high level of acetic acid) for methane production and growth existed in both anaerobic reactors (Table 4), many bands disappeared, and major band intensities (A7 and A16) decreased with 10-d SRT operation. This phenomenon was explained by the methanogens being susceptible to the pH variation (Jun et al., 2009). Also, a relatively short SRT partially led to the washout of methanogens in both anaerobic reactors.

Collectively, the bacteria and archaea DGGE profiles indicated that TAD pretreatment led to a higher diversity of the bacteria community in $R_{MAD,}$ resulting in enhanced hydrolysis and methane production compared to $R_{CONTROL}$ at SRTs of 40 and 20 days. However, with a relatively short SRT of 10 days, the archaea community diversities in R_{MAD} and $R_{CONTROL}$ decreased markedly.

3.5.2. *Quantitative analysis by qPCR*

To quantitatively investigate the microbial population change and diversity in relation to reactor performance under different SRTs, qPCR was performed, targeting the total bacteria, archaea, and the four major methanogens (Table 3). As shown in Fig. 6a, R_{TAD} maintained the highest bacteria population in terms of 16S rRNA gene concentration (9.28×10^{10}–1.05×10^{11} copies/mL) during the overall digestion, although it has a relatively short SRT (1 day). This notable maintenance of a higher bacteria population than R_{MAD} and $R_{CONTROL}$ seems to be due to the aerobic microbes having relatively faster growth behavior than anaerobic microbes. Also, it might be partly attributed to the high bacteria concentration in the seed sludge obtained from the ATAD pilot process. During the overall digestion, R_{MAD} usually showed higher bacteria 16S rRNA gene concentration than $R_{CONTROL}$, since R_{TAD} may accelerate hydrolysis and produce sufficient SOM. The concentration of bacteria 16S rRNA gene concentration in both anaerobic reactors increased with decreasing SRT until 20-d operation. Its population increased to 6.73×10^{10} copies/mL (a 2.11-fold increase compared to the seed sludge) in $R_{MAD,}$ and to 4.59×10^{10} copies/mL (a 1.5-fold increase compared to the seed sludge) in $R_{CONTROL}$ at the 20-d SRT, respectively. On the other hand, a significant decline of the bacteria population in the R_{MAD} and $R_{CONTROL}$ was observed with 10-d SRT operation. These results implied that insufficient SRT was correlated with the deterioration of reactor stability (Fig. 2).

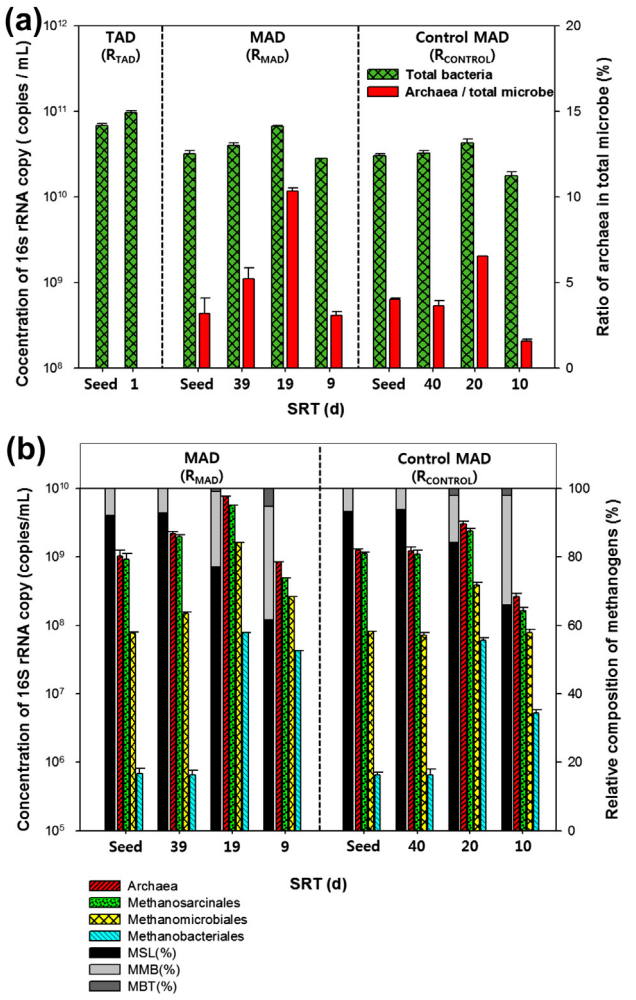


Fig. 6 – Change in 16S rRNA concentration of (a) total bacteria and ratio of archaea in total microbe, (b) archaea, four major methanogens and relative composition of methanogens with the SRTs.

The 16S rRNA gene concentration of archaea and the four methanogens with respect to SRTs is visualized in Fig. 6b. Theoretically, the sum of 16S rRNA gene concentrations of the four methanogens quantified with the specific primer is equal to that of archaea. The ratios of the sum of 16S rRNA gene concentrations of the four methanogens to that of archaea were determined as 0.95 and 0.96 in R_{MAD} and $R_{CONTROL}$ during the overall digestion, respectively. All primers used in this study were highly specific to the targeted microorganisms. As shown in Fig. 6b, the archaea population increased to 7.75×10^9 copies/mL (a 7.5-fold increase relative to the seed sludge) in R_{MAD} and 3.02×10^9 copies/mL (a 2.4-fold increase relative to the seed sludge) in $R_{CONTROL}$ with the 20-d SRT, respectively. Interestingly, this marked increment was closely linked to the increment of bacteria (Fig. 6a) and the significant

increment in MPR in R_{MAD} and $R_{CONTROL}$ (Fig. 4a). In addition, the proportion of archaea to the total microbe amount (the sum of archaea and bacteria) in R_{MAD} and $R_{CONTROL}$ shows the highest values of 10.5 and 6.5% with 20-d SRT operation, respectively (Fig. 6a). Typically, these values are considered as indirect indicators that are highly linked to methane production and content (Raskin et al., 1995). Our results indicate that the increment of this proportion corresponded to the increment of MPR and methane content in R_{MAD} and $R_{CONTROL}$ (Fig. 4).

The changes in the 16S rRNA gene concentration and relative composition of the order-level methanogens with respect to SRTs are shown in Fig. 6b. In the seed sludge of R_{MAD} and $R_{CONTROL}$, the aceticlastic methanogen order MSL accounts for 92.2 and 93.8% of the archaea population, respectively. The orders MMB and MBT belonging to the hydrogenotrophic methanogen population represented a low proportion (<7% of archaea) of the population in the initial digestion period. As the SRT decreased to 20 days, the relative composition of hydrogenotrophic methanogens in R_{MAD} and $R_{CONTROL}$, including MMB and MBT, increased to 22.8 and 15.8% of the archaea populations, respectively. The populations of order MMB in R_{MAD} increased to 1.63×10^9 copies/mL (a 21-fold increase relative to the seed sludge). This marked increment of the hydrogenotrophic methanogen population with the 20-d SRT operation indicate that (1) a relatively short SRT is more favorable for hydrogenotrophic methanogens than aceticlastic methanogens, because its growth rate is higher (Bonin and Boone, 2007), and (2) a partial increase of OLR might lead to the enrichment of syntrophic VFA oxidizing bacteria (e.g., *Clostridium* sp.), which are highly coupled with hydrogenotrophic methanogens (Hattori, 2008). It has already been reported that some part of VFA, especially acetate, can be converted to hydrogen and CO_2 during the relatively short HRT operation, although acetate was directly converted to methane by the aceticlastic methanogens (Westerholm et al., 2010; Sasaki et al., 2011).

As with the archaea DGGE profiles, a significant decline in the methanogen population with 10-d SRT operation was observed in R_{MAD} and $R_{CONTROL}$. This result was consistent with previous studies which reported that overloading and short retention time lead to the accumulation of VFA and an abrupt decline in pH, which can cause a decrease in methanogenic activity and population (Chen et al., 2008; Ma et al., 2009). Overall, the reactor performance was consistent with the qPCR results during the digestion. 10-d SRT was not sufficient to maintain sufficient populations of bacteria and methanogens. Consequently, MAD with TAD should be operated with SRTs of 20 days or more to prevent the washout of bacteria and methanogens, which leads to the deterioration of reactor performance. Collectively, the experimental results demonstrated that the increased COD removal and methane production with different SRTs in R_{MAD} might be attributed to the increased synergism among the bacteria and methanogen species by improving the hydrolysis of rate limiting step in the sludge with the help of the biological TAD pre-treatment. Although this study has considered microbial community structure dynamics with different SRTs, further quantification of the individual bacteria species such as syntrophic VFA oxidizing bacteria will be required for better understanding of combined process.

4. Conclusions

This study has focused on the applicability of the TAD process as a sludge pre-treatment step, and the response of qualitative and quantitative microbial community to SRT changes of mesophilic anaerobic digestion resulting in variations of sludge reduction and methane production in a sequential TAD-MAD process. Even with a 1-day SRT, R_{TAD} maintains high stability and solid removal efficiency (27% of VSSR) during the overall digestion. The sequential TAD-MAD process shows higher MPR by about 42%, and a 15% higher TCOD removal than the control MAD during the operation with a 20-d SRT. This difference in performance between the $R_{CONTROL}$ and the combined process demonstrates the advantages of TAD as a pre-treatment stage in the sludge digestion process. The bacteria DGGE profiles in R_{TAD} showed that thermophilic *Bacillus* spp. and cellulolytic *Clostridia* spp. were mainly detected and highly responsible for WAS degradation. When the TAD process was applied for biological pre-treatment, additional bacteria bands involved in hydrolysis and acidogenesis were detected in R_{MAD}. The analysis of the archaea community structures indicated that Methanosacinales showed the highest population in both anaerobic reactors during all SRTs, which corresponds to *Methanosaeta concilli*, a strict aceticlastic methanogen. The presence at 20-and 10-d SRT of *C. ultunense* sp., syntrophic VFA oxidizing bacteria, along with *M. beijingense* sp., a hydrogenotrophic methanogen, suggests their contribution to hydrogenotrophic methanogenesis. Also, marked increment of hydrogenotrophic methanogen species including MMB and MBT as the SRT decreased from 40 to 20 days in R_{MAD} and $R_{CONTROL}$ indicated that the hydrogenotrophic methanogen species might be affected more by SRT than the aceticlastic methanogen species.

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REFERENCES

Abeynayaka, A., Visvanathan, C., 2011. Performance comparison of mesophilic and thermophilic aerobic sidestream membrane bioreactors treating high strength wastewater. Bioresource Technology 102 (9), 5345–5352.

Abubaker, J., Risberg, K., Pell, M., 2012. Biogas residues as fertilisers – effects on wheat growth and soil microbial activities. Applied Energy 99, 126–134.
Anson, M.L., 1938. The estimation of Pepsin, Trypsin, Papain, and Cathepsin with Hemoglobin. The Journal of General Physiology 22 (1), 79–89.
APHA–AWWA–WEF, 1998. Standard Methods for the Examination of Water and Wastewater, 20th ed. American Public Health Association, Washington, D.C.
Appels, L., Baeyens, J., Degrève, J., Dewil, R., 2008. Principles and potential of the anaerobic digestion of waste-activated sludge. Progress in Energy and Combustion Science 34 (6), 755–781.
Björnsson, L., Murto, M., Jantsch, T.G., Mattiasson, B., 2001. Evaluation of new methods for the monitoring of alkalinity, dissolved hydrogen and the microbial community in anaerobic digestion. Water Research 35 (12), 2833–2840.
Bonin, A., Boone, D., 2007. The order methanobacteriales. In: Dworkin, M., Falkow, S., Rosenberg, E., Schleifer, K.-H., Stackebrandt, E. (Eds.), The Prokaryotes. Springer, New York, pp. 231–243.
Campbell, H.W., 2000. Sludge management – future issues and trends. Water Science & Technology 41 (8), 1–8.
Carrère, H., Dumas, C., Battimelli, A., Batstone, D.J., Delgenès, J.P., Steyer, J.P., Ferrer, I., 2010. Pretreatment methods to improve sludge anaerobic degradability: a review. Journal of Hazardous Materials 183 (1–3), 1–15.
Chen, X.-G., Stabnikova, O., Tay, J.-H., Wang, J.-Y., Tay, S.-L., 2004. Thermoactive extracellular proteases of Geobacillus caldoproteolyticus, sp. nov., from sewage sludge. Extremophiles 8 (6), 489–498.
Chen, Y., Cheng, J.J., Creamer, K.S., 2008. Inhibition of anaerobic digestion process: a review. Bioresource Technology 99 (10), 4044–4064.
Coelho, N.M.G., Droste, R.L., Kennedy, K.J., 2011. Evaluation of continuous mesophilic, thermophilic and temperature phased anaerobic digestion of microwaved activated sludge. Water Research 45 (9), 2822–2834.
Cupp-Enyard, C., 2008. Sigma's non-specific protease activity assay – casein as a substrate. Journal of Visualized Experiments 19, e899. http://dx.doi.org/10.3791/899.
DuBois, M., Gilles, K.A., Hamilton, J.K., Rebers, P.A., Smith, F., 1956. Colorimetric method for determination of sugars and related substances. Analytical Chemistry 28 (3), 350–356.
Fang, H., Yu, H., 2000. Effect of HRT on mesophilic acidogenesis of dairy wastewater. Journal of Environmental Engineering 126 (12), 1145–1148.
Gao, D.W., Tao, Y., 2012. Current molecular biologic techniques for characterizing environmental microbial community. Frontiers of Environmental Science & Engineering 6 (1), 82–97.
Ge, H., Jensen, P.D., Batstone, D.J., 2011. Temperature phased anaerobic digestion increases apparent hydrolysis rate for waste activated sludge. Water Research 45 (4), 1597–1606.
Gomez, J., de Gracia, M., Ayesa, E., Garcia-Heras, J.L., 2007. Mathematical modelling of autothermal thermophilic aerobic digesters. Water Research 41 (5), 959–968.
Gujer, W., Zehnder, A.J.B., 1983. Conversion processes in anaerobic digestion. Water Science and Technology 15 (8–9), 127–167.
Han, Y., Sung, S., Dague, R.R., 1997. Temperature-phased anaerobic digestion of wastewater sludges. Water Science and Technology 36 (6–7), 367–374.
Hasegawa, S., Shiota, N., Katsura, K., Akashi, A., 2000. Solubilization of organic sludge by thermophilic aerobic bacteria as a pretreatment for anaerobic digestion. Water Science and Technology 41 (3), 163–169.
Hattori, S., 2008. Syntrophic acetate-oxidizing microbes in methanogenic environments. Microbes and Environments 23 (2), 118–127.
Jun, D., Yong-sheng, Z., Mei, H., Wei-hong, Z., 2009. Influence of alkalinity on the stabilization of municipal solid waste in anaerobic simulated bioreactor. Journal of Hazardous Materials 163 (2–3), 717–722.
Kato, S., Haruta, S., Cui, Z.J., Ishii, M., Yokota, A., Igarashi, Y., 2004. Clostridium straminisolvens sp. nov., a moderately thermophilic, aerotolerant and cellulolytic bacterium isolated from a cellulose-degrading bacterial community. International Journal of Systematic and Evolutionary Microbiology 54 (6), 2043–2047.
Keffala, C., Harerimana, C., Vasel, J.-l, 2013. A review of the sustainable value and disposal techniques, wastewater stabilisation ponds sludge characteristics and accumulation. Environmental Monitoring and Assessment 185 (1), 45–58.
Kumar, N., Novak, J.T., Water, D.C., 2006. sequential anaerobic-aerobic digestion for enhanced volatile solids reduction and nitrogen removal. Proceedings of the Water Environment Federation 2006 (2), 1064–1081.
LaPara, T.M., Alleman, J.E., 1999. Thermophilic aerobic biological wastewater treatment. Water Research 33 (4), 895–908.
Lee, C., Kim, J., Shin, S.G., Hwang, S., 2008. Monitoring bacterial and archaeal community shifts in a mesophilic anaerobic batch reactor treating a high-strength organic wastewater. FEMS Microbiology Ecology 65 (3), 544–554.
Lee, M., Hidaka, T., Hagiwara, W., Tsuno, H., 2009. Comparative performance and microbial diversity of hyperthermophilic and thermophilic co-digestion of kitchen garbage and excess sludge. Bioresource Technology 100 (2), 578–585.
Levén, L., Eriksson, A.R.B., Schnürer, A., 2007. Effect of process temperature on bacterial and archaeal communities in two methanogenic bioreactors treating organic household waste. FEMS Microbiology Ecology 59 (3), 683–693.
Li, Y., Park, S.Y., Zhu, J., 2011. Solid-state anaerobic digestion for methane production from organic waste. Renewable and Sustainable Energy Reviews 15 (1), 821–826.
Liu, S., Song, F., Zhu, N., Yuan, H., Cheng, J., 2010. Chemical and microbial changes during autothermal thermophilic aerobic digestion (ATAD) of sewage sludge. Bioresource Technology 101 (24), 9438–9444.
Lowry, O.H., Rosebrough, N.J., Lewis Farr, A., Randall, R.J., 1951. Protein measurement with the Folin phenol reagent. Journal of Biological Chemistry 193, 265–275.
Ma, J., Mungoni, L.J., Verstraete, W., Carballa, M., 2009. Maximum removal rate of propionic acid as a sole carbon source in UASB reactors and the importance of the macro- and micro-nutrients stimulation. Bioresource Technology 100 (14), 3477–3482.
McInerney, M.J., 1988. Anaerobic hydrolysis of fats and proteins. In: Zehnder, A.J.B. (Ed.), Biology of Anaerobic Micro-organisms. John Wiley and Sons, New York, pp. 373–415.
MOE, 2012. Sewage Sludge Management and Planning (Seoul, Korea).
Nakamura, L.K., 1981. Lactobacillus amylovorus, a new starch-hydrolyzing species from cattle waste-corn fermentations. International Journal of Systematic Bacteriology 31 (1), 56–63.
Navia, R., Soto, M., Vidal, G., Bornhardt, C., Diez, M.C., 2002. Alkaline pretreatment of kraft mill sludge to improve its anaerobic digestion. Bulletin of Environmental Contamination and Toxicology 69 (6), 0869–0876.
Nocker, A., Burr, M., Camper, A.K., 2007. Genotypic microbial community profiling: a critical technical review. Microbial Ecology 54 (2), 276–289.
Novak, J.T., Banjade, S., Murthy, S.N., 2011. Combined anaerobic and aerobic digestion for increased solids reduction and nitrogen removal. Water Research 45 (2), 618–624.
Pagilla, K.R., Kim, H., Cheunbarn, T., 2000. Aerobic thermophilic and anaerobic mesophilic treatment of swine waste. Water Research 34 (10), 2747–2753.

Paul, E., Camacho, P., Sperandio, M., Ginestet, P., 2006. Technical and economical evaluation of a thermal, and two oxidative techniques for the reduction of excess sludge production. Process Safety and Environmental Protection 84 (4), 247–252.

Piterina, A.V., Bartlett, J., Tony Pembroke, J., 2012. Phylogenetic analysis of the bacterial community in a full scale autothermal thermophilic aerobic digester (ATAD) treating mixed domestic wastewater sludge for land spread. Water Research 46 (8), 2488–2504.

Raskin, L., Zheng, D., Griffin, M.E., Stroot, P.G., Misra, P., 1995. Characterization of microbial communities in anaerobic bioreactors using molecular probes. Antonie van Leeuwenhoek. International Journal of General and Molecular Microbiology 68 (4), 297–308.

Rintala, J.A., Puhakka, J.A., 1994. Anaerobic treatment in pulp- and paper-mill waste management: a review. Bioresource Technology 47 (1), 1–18.

Salinas, M.B., Fardeau, M.-L., Cayol, J.-L., Casalot, L., Patel, B.K.C., Thomas, P., Garcia, J.-L., Ollivier, B., 2004. Petrobacter succinatimandens gen. nov., sp. nov., a moderately thermophilic, nitrate-reducing bacterium isolated from an Australian oil well. International Journal of Systematic and Evolutionary Microbiology 54 (3), 645–649.

Sasaki, D., Hori, T., Haruta, S., Ueno, Y., Ishii, M., Igarashi, Y., 2011. Methanogenic pathway and community structure in a thermophilic anaerobic digestion process of organic solid waste. Journal of Bioscience and Bioengineering 111 (1), 41–46.

Schmit, K.H., Ellis, T.G., 2001. Comparison of temperature-phased and two-phase anaerobic co-digestion of primary sludge and municipal solid waste. Water Environment Research 73 (3), 314–321.

Schnürer, A., Schink, B., Svensson, B.H., 1996. Clostridium ultunense sp. nov., a mesophilic bacterium oxidizing acetate in syntrophic association with a hydrogenotrophic methanogenic bacterium. International Journal of Systematic Bacteriology 46 (4), 1145–1152.

Shin, S.G., Han, G., Lim, J., Lee, C., Hwang, S., 2010a. A comprehensive microbial insight into two-stage anaerobic digestion of food waste-recycling wastewater. Water Research 44 (17), 4838–4849.

Shin, S.G., Lee, S., Lee, C., Hwang, K., Hwang, S., 2010b. Qualitative and quantitative assessment of microbial community in batch anaerobic digestion of secondary sludge. Bioresource Technology 101 (24), 9461–9470.

Simpson, P.J., Ross, R.P., Fitzgerald, G.F., Stanton, C., 2004. Bifidobacterium psychraerophilum sp. nov. and Aeriscardovia aeriphila gen. nov., sp. nov., isolated from a porcine caecum. International Journal of Systematic and Evolutionary Microbiology 54 (2), 401–406.

Song, Y.-C., Kwon, S.-J., Woo, J.-H., 2004. Mesophilic and thermophilic temperature co-phase anaerobic digestion compared with single-stage mesophilic- and thermophilic digestion of sewage sludge. Water Research 38 (7), 1653–1662.

Stuckey, D.C., McCarty, P.L., 1984. The effect of thermal pretreatment on the anaerobic biodegradability and toxicity of waste activated sludge. Water Research 18 (11), 1343–1353.

Supaphol, S., Jenkins, S.N., Intomo, P., Waite, I.S., O'Donnell, A.G., 2011. Microbial community dynamics in mesophilic anaerobic co-digestion of mixed waste. Bioresource Technology 102 (5), 4021–4027.

Tomei, M.C., Rita, S., Mininni, G., 2011. Performance of sequential anaerobic/aerobic digestion applied to municipal sewage sludge. Journal of Environmental Management 92 (7), 1867–1873.

Toreci, I., Kennedy, K.J., Droste, R.L., 2009. Evaluation of continuous mesophilic anaerobic sludge digestion after high temperature microwave pretreatment. Water Research 43 (5), 1273–1284.

Ueda, K., Yamashita, A., Ishikawa, J., Shimada, M., Watsuji, T.-O., Morimura, K., Ikeda, H., Hattori, M., Beppu, T., 2004. Genome sequence of Symbiobacterium thermophilum, an uncultivable bacterium that depends on microbial commensalism. Nucleic Acids Research 32 (16), 4937–4944.

Ugwuanyi, J.O., Harvey, L.M., McNeil, B., 2008. Diversity of thermophilic populations during thermophilic aerobic digestion of potato peel slurry. Journal of Applied Microbiology 104 (1), 79–90.

Vavilin, V.A., Fernandez, B., Palatsi, J., Flotats, X., 2008. Hydrolysis kinetics in anaerobic degradation of particulate organic material: an overview. Waste Management 28 (6), 939–951.

Ward, A.J., Hobbs, P.J., Holliman, P.J., Jones, D.L., 2008. Optimisation of the anaerobic digestion of agricultural resources. Bioresource Technology 99 (17), 7928–7940.

Weemaes, M.P.J., Verstraete, W.H., 1998. Evaluation of current wet sludge disintegration techniques. Journal of Chemical Technology & Biotechnology 73 (2), 83–92.

Westerholm, M., Roos, S., Schnürer, A., 2010. Syntrophaceticus schinkiigen. nov., sp. nov., an anaerobic, syntrophic acetate-oxidizing bacterium isolated from a mesophilic anaerobic filter. FEMS Microbiology Letters 309 (1), 100–104.

Whelan, J.A., Russell, N.B., Whelan, M.A., 2003. A method for the absolute quantification of cDNA using real-time PCR. Journal of Immunological Methods 278 (1–2), 261–269.

Yan, S., Miyanaga, K., Xing, X.-H., Tanji, Y., 2008. Succession of bacterial community and enzymatic activities of activated sludge by heat-treatment for reduction of excess sludge. Biochemical Engineering Journal 39 (3), 598–603.

Yang, Q., Luo, K., Li, X.-M., Wang, D.-B., Zheng, W., Zeng, G.-M., Liu, J.-J., 2010. Enhanced efficiency of biological excess sludge hydrolysis under anaerobic digestion by additional enzymes. Bioresource Technology 101 (9), 2924–2930.

Ziemba, C., Peccia, J., 2011. Net energy production associated with pathogen inactivation during mesophilic and thermophilic anaerobic digestion of sewage sludge. Water Research 45 (16), 4758–4768.